

solution 溶液
solute 溶质
solvent 溶剂
dispersed 分散

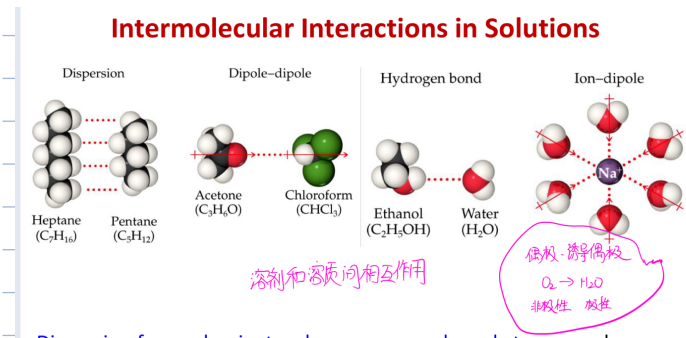
① solvate 溶剂化

≠ 溶解全过程

让溶质分子分开

solvation

② 溶剂和溶质间的相互作用



③ How Does a solution form

If an ionic salt is soluble in water, it is because the ion-dipole interactions are strong enough to overcome the **lattice energy** of the salt crystal.

晶格能

解释为什么吸热溶解能发生

包含了三个过程

(b) The increase of entropy favors the formation of the solution.

$$\Delta H_{\text{soln}} = \Delta H_{\text{solute}} + \Delta H_{\text{solvent}} + \Delta H_{\text{mix}}$$

→ 一定

- $\Delta H_{\text{solute}} > 0$ (endothermic), require energy to overcome attraction force between solute-solute interactions.
- $\Delta H_{\text{solvent}} > 0$ (endothermic), require energy to overcome attraction force between solvent-solvent interactions.
- $\Delta H_{\text{mix}} < 0$ (exothermic), energy given out to allow attraction between solvent and solvent molecule.

if $\Delta H_{\text{soln}} > 0$, endothermic

if $\Delta H_{\text{soln}} < 0$, exothermic

经典案例 NH_4NO_3

$MgSO_4$

Dissolution 溶解 不是反应 → 物理过程 (看能否 go back)

④ Types of solution

saturated 饱和 那个温度下已经最大程度, 处于动态平衡, 溶解(结晶)同时发生, 暗示有溶质沉淀

unsaturated 不饱和

supersaturated 过饱和 unstable stimulated 刺激

⑤ 影响溶解度因素

1° like dissolves like

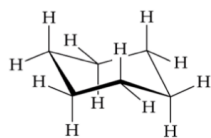
Polar → Polar

Nonpolar → Nonpolar

The more similar the intermolecular attractions, the more likely one substance is to be soluble in another.

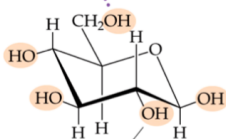
葡萄糖

Glucose (which has hydrogen bonding) is very soluble in water, while cyclohexane (which only has dispersion forces) is not.



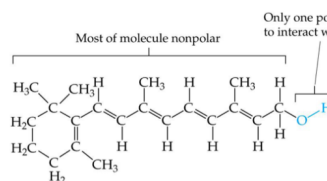
Cyclohexane, C_6H_{12} , which has no polar OH groups, is essentially insoluble in water

OH groups enhance the aqueous solubility because of their ability to hydrogen bond with H_2O .

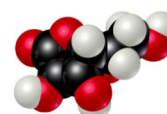
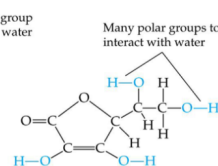


Glucose, $C_6H_{12}O_6$, has five OH groups and is highly soluble in water

- Vitamin A is soluble in nonpolar compounds (like fats).
- Vitamin C is soluble in water.



Vitamin A



Vitamin C

2° 气体

TABLE 13.1 • Solubilities of Gases in Water at 20 °C, with 1 atm Gas Pressure

Gas	Solubility (M)
N_2	0.69×10^{-3}
CO	1.04×10^{-3}
O_2	1.38×10^{-3}
Ar	1.50×10^{-3}
Kr	2.79×10^{-3}

- Larger molecules have stronger dispersion forces.

(b) As molar mass and atomic size of the ideal gas increases, the electron cloud is more easily polarizable. The greater the polarizability, the stronger the dispersion forces between the gas atoms and water, leading to greater solubility of the atom.

气压 ← ☆☆

固体、液体溶解度不受气压影响

气体受，呈 directly proportional 直接正比

⑥ Henry's law

气压 ≠ 蒸气压

$$S_g = kP_g$$

the solubility of a gas in a liquid solvent increases in direct proportion to the partial pressure of the gas above the solution

亨利定律(Henry's law): "在等温等压下, 某种气体在溶液中的溶解度与液面上该气体的平衡压力成正比。" 这一定律对于稀溶液中挥发性溶质也同样有用。

$$S_g = kP_g$$

where

- S_g is the solubility of the gas (unit: molarity),
- k is the Henry's Law constant for that gas in that solvent, and
- P_g is the partial pressure of the gas above the liquid.



挥发性溶质 ≈ 气体

针对某一溶质某一溶剂

温度 ← ★★

对固体 $T \uparrow S \uparrow$ (有特例)

对气体 $T \uparrow S \downarrow$

⑦ 表达

Mass Percentage (质量百分比)

$$\text{Mass \% of A} = \frac{\text{mass of A in solution}}{\text{total mass of solution}} \times 100\%$$



Parts per million (ppm)

$$\text{ppm} = \frac{\text{mass of A in solution}}{\text{total mass of solution}} \times 10^6$$

Parts per billion (ppb)

$$\text{ppb} = \frac{\text{mass of A in solution}}{\text{total mass of solution}} \times 10^9$$

Mole Fraction (X) (摩尔分数)

$$X_A = \frac{\text{moles of A}}{\text{total moles of all components}}$$

- In some applications, one needs the mole fraction of *solvent*, not solute—make sure you find the quantity you need!

Molarity (M)

$$M = \frac{\text{moles of solute}}{\text{liters of solution}}$$

Molality (m)

$$m = \frac{\text{moles of solute}}{\text{kilograms of solvent}}$$

- Molarity (M) depends on *volume of solution*
 - volume is temperature-dependent, therefore molarity can change with temperature.
 - Molality (m) depends on the *mass of solvent*
 - both moles and mass do not change with temperature, therefore molality is temperature-independent.
- 当涉及温度变化时, 常用重量摩尔浓度 molality

⑧ Colligative Properties 依数性

只和溶质量有关, 与溶质性质无关

1° Vapor Pressure

溶剂蒸气压下降

加入了 nonvolatile solutes
不易挥发

气体
气压 ≠ 蒸气压
溶剂

Raoult's Law 拉乌尔定律

$$P_A = X_A P_A^\circ$$

where

- X_A is the mole fraction of solvent A, and
- P_A° is the normal vapor pressure of solvent A at that temperature. 纯溶剂

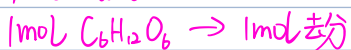
Note: This is one of those times when you want to make sure you have the vapor pressure of the *solvent* (only the solvent is volatile!).

Colligative Properties (依数性)

- Addition of solute into a solvent would **lower freezing point** and **raise boiling point**
- These properties depends only on the **quantity** (number) of solute particles present, **not on the identity** of the solute particles, called **colligative properties**.
稀溶液中溶剂的蒸气压下降、凝固点降低、沸点升高和渗透压的数值, 只与溶液中溶质的量有关, 与溶质的本性无关, 故称这些性质为稀溶液的依数性 (溶液的通性)
- Among colligative properties are
 - Lowering of vapor-pressure
 - Raising of boiling-point
 - Lowering of melting-point
 - Osmotic pressure 渗透压

★ 注意加的是什么

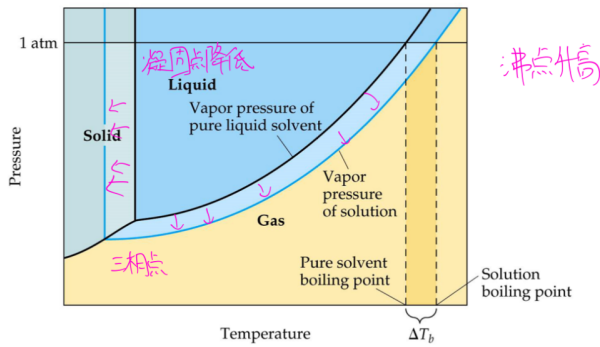
离子化合物和分子可不一样



2° Boiling-point elevation 升高

Melting-point depression 降低

Boiling-Point Elevation and Freezing-Point Depression



Nonvolatile solute-solvent interactions also cause solutions to have higher boiling points and lower freezing points than the pure solvent.

$$\Delta T_b = k_b \cdot m \cdot i$$

$$\Delta T_f = k_f \cdot m \cdot i$$

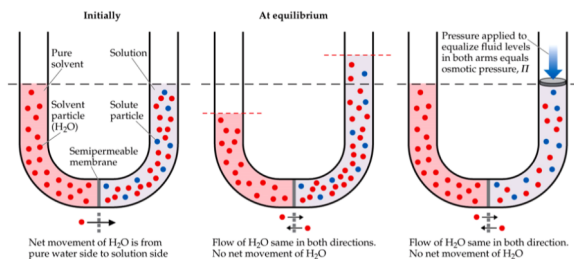
小m
针对离子化合物电解质
电解质
跟溶剂有关
是否理想看情况
越稀越理想

3° Osmosis 渗透

Semipermeable membranes 半透膜

基本是允许水过

Osmosis (渗透)



In osmosis, there is net movement of solvent from the area of **higher solvent concentration (lower solute concentration)** to the area of **lower solvent concentration (higher solute concentration)**.

从低溶质浓度 → 高溶质浓度

Osmotic Pressure

The pressure required to stop osmosis, known as **osmotic pressure** 渗透压, Π , is

$$\Pi = \left(\frac{n}{V}\right)RT = MRT$$

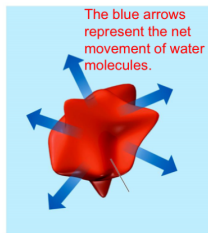
★ 电解质还是有系数影响

where M is the **molarity** of the solution and R is ideal-gas constant.

If the osmotic pressure is the same on both sides of a membrane (i.e., the concentrations are the same), the solutions are **isotonic** 等渗的

Osmosis in Blood Cells

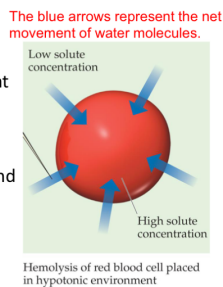
- If the **solute concentration** outside the cell is **greater than** that **inside** the cell, the solution is **hypertonic** (高渗的).
- Water will **flow out** of the cell, and **crenation** (皱缩) results.



高低针对 cell 处环境渗透压

Osmosis in Blood Cells

- If the **solute concentration** outside the cell is **less** than that inside the cell, the solution is **hypotonic** (低渗的).
- Water will **flow into** the cell, and **hemolysis** (溶血) results.



The average osmotic pressure of blood is 7.7 atm at 25 °C.

Reverse Osmosis 反渗透
需要外加能量

9 Colloids 胶体

Colloids (胶体)

Suspensions (悬浮) of **particles larger than** individual **ions or molecules**, but **too small to be settled out** by gravity, are called **colloids**.

Colloid particles range in diameter from 5 to 1000 nm

TABLE 13.5 • Types of Colloids

Phase of Colloid	Dispersing (solvent-like) Substance	Dispersed (solute-like) Substance	Colloid Type	Example
Gas	Gas	Gas	—	None (all are solutions)
Gas	Gas	Liquid	Aerosol	Fog
Gas	Gas	Solid	Aerosol	Smoke
Liquid	Liquid	Gas	Foam	Whipped cream
Liquid	Liquid	Liquid	Emulsion	Milk
Liquid	Liquid	Solid	Sol	Paint
Solid	Solid	Gas	Solid foam	Marshmallow
Solid	Solid	Liquid	Solid emulsion	Butter
Solid	Solid	Solid	Solid sol	Ruby glass

Tyndall Effect (丁达尔效应)

solution **Colloidal dispersion**



- Colloidal suspensions can **scatter rays of light**.
- This phenomenon is known as the **Tyndall effect**.

丁达尔效应:
当一束光线透过胶体，从入射光的垂直方向可以观察到胶体里出现的一条光亮的“通路”，这种现象叫 (Tyndall effect)



hydrophilic

亲水

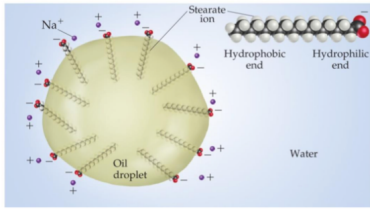
hydrophobic

疏水

Emulsifying agents 乳化剂

Some molecules have a **polar, hydrophilic** (*water-loving*) end and a **nonpolar, hydrophobic** (*water-hating*) end. **Hydrophobic** groups are **away from water**, on the "inside" of molecules.

These molecules can **help in the emulsification** (乳化) of **fats and oils** in **aqueous** solutions.



Soap: $C_{17}H_{35}COONa$

removal < $\begin{matrix} \text{heat} \\ \text{coagulation} \\ \text{Dialysis} \end{matrix}$ $\begin{matrix} \text{聚沉} \\ \text{渗析} \end{matrix}$

miscible 能混溶的

adsorption 吸附

$$1\text{cm}^3 = 1\text{mL}$$